

Incorporate precisely engineered microstructures on intraocular lens surface that manipulate light patterns to promote natural tear film distribution and meibomian gland function

Executive Technology Brief

Domain: optical_photonic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'adverse_event_without_identified_device_' for intraocular lens: 17 MAUDE reports across 4 distinct devices (20200512 to 20201224); root-cause evidence from 14 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 14 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cus Unresolved gap: Across the 5 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, event, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, producing, recurring, reported, unde

Mechanism: Passive light manipulation through microstructured surface to enhance tear film stability without active pharmaceutical intervention

Architecture: PROPOSED architecture (no physical system exists): source (emitter / external illuminator); transport medium (free-space / fiber / tissue); receiver / conversion element; thermal path and safety interlocks — realizing: Incorporate precisely engineered microstructures on intraocular lens surface that manipulate light patterns to promote natural tear film distribution and meibomian gland function

Governing Model: Optical transport / photonic power governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

OPT-003: Numerical aperture. Expression: $NA = n * \sin(\theta)$. Applicability to this invention: CONDITIONAL; holds under: paraxial regime; homogeneous medium at the interface. Assumptions: coupling/acceptance angles follow from NA and must be verified on the actual geometry. Engaged variables: NOT ESTABLISHED.

OPT-004: Thermal load deposited at target. Expression: $Q = I * A - P_{conv}$. Applicability to this invention: CONDITIONAL; holds under: steady state; unconverted power deposits as heat. Assumptions: steady temperature requires a conduction/perfusion model before any quantitative claim; P_{conv} is the converted output power; the conversion fraction $\eta_{conversion}$ is MODEL_DERIVED until measured. Engaged variables: NOT ESTABLISHED.

OPT-001: Beer-Lambert attenuation. Expression: $I = I_0 * \exp(-\mu_a * d)$. Applicability to this invention: CONDITIONAL; holds under: homogeneous medium; effective coefficient folds scattering into μ_a . Assumptions: tissue is heterogeneous; μ_a is a MODEL_DERIVED effective value. Engaged variables: NOT ESTABLISHED.

OPT-002: Photovoltaic conversion. Expression: $P_{elec} = \eta * I * A$. Applicability to this invention: CONDITIONAL; holds under: cell operated below saturation; η at actual wavelength and temperature. Assumptions: η under monochromatic NIR differs from solar-rated η . Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.